ELECTROCHEMISTRY

Electrochemistry is the branch of Physical chemistry which deals with the study of conversion of

* chemical energy into electrical energy (Galvanic cell)
* Electrical energy into chemical energy (Electrolytic cell)

PART- ONE - GENERAL POINTS

1. CONDUCTANCE

Those substances which can conduct electricity are called Conductors. Conductors are of two types

* Metallic Conductors (electronic Conductors)
* Electrolytic Conductors (Ionic Conductors)

] BOTH FOLLOW OHM'S LAW

METALLIC CONDUCTOR	ELECTROLYTIC CONDUCTOR
Flow of current due to movement of free electrons	Flow of current due to movement of ions
Doesnot involves transfer of matters	Involves transfer of matters
Does not Involve any chemical changes	Involves chemical changes
Conducting Power is usually very High	Conducting Power is comparatively less
Doesnot follow faraday's law of electrolysis	follows faradays' Laws of electrolysis
On increasing Temperature Resistance increases Conductivity decreases	On Increasing Temperature Resistance decreases Conductivity increases

CLASSNOTES
KOTA

2. CONDUCTANCE TERMS

2.A- CONDUCTANCE [C / G]

NCERT ने conductance के लिए [G] use किया हुआ है।

Reciprocal of Resistance is called as conductance

$$G = \frac{1}{R}$$

Unit = $\Omega^{-1} = ohm^{-1} = mho$ = siemens (S)

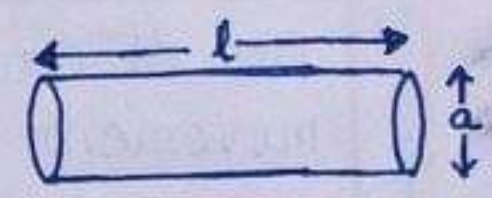
2.B- SPECIFIC CONDUCTANCE / CONDUCTIVITY [σ / K]

PHYSICS USES SIGN = σ
CHEMISTRY USES SIGN = K

IN PHYSICS CONCEPT OF [σ]

$$R = \rho \times \frac{l}{a} \Rightarrow \rho = \frac{Ra}{l}$$

↓
SPECIFIC RESISTANCE
OR
RESISTIVITY



Reciprocal of Resistivity is called Conductivity
OR
SPECIFIC RESISTANCE
OR
SPECIFIC CONDUCTANCE

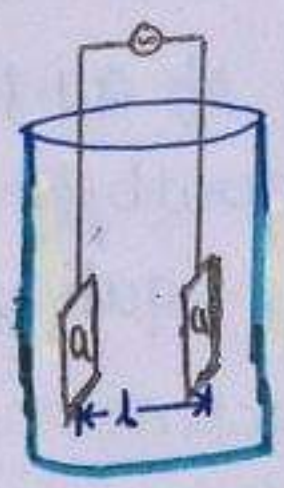
$$\sigma = \frac{1}{\rho}$$

$$\boxed{\sigma = \frac{1}{R} \times \frac{l}{a}}$$

unit = ρ (Resistivity) = Ωm OR Ωcm

σ (conductivity) = $\Omega^{-1} m^{-1}$ OR $\Omega^{-1} cm^{-1}$ OR Sm^{-1} OR Scm^{-1}

IN CHEMISTRY CONCEPT OF [K]



* l = distance between electrode (generally in cm)

Battery कोई metre या दो मीटर का generally नही होता है।

* a = Dipped surface area

ये surface area लेते वक्त दिमाग और आँख का बत्ती जला लो वरना सामने wrong answer खड़ा है।

* (l) and (a) donot depends upon nature of electrolysis.
* for a given cell (l) and (a) always remain same

therefore $\frac{l}{a}$ is also called cell constant [C*/G*]

Reason ये है कि Battery का size एक बार जो बन गया सो बन गया अब Battery थोड़े ना अपना length और area increase या decrease करेगा thats why $\frac{l}{a}$ is constant and called cell constant [C*/G*]

$$K = \frac{1}{R} \cdot \frac{l}{a}$$

$$\boxed{K = G \times G^*}$$

K = conductance x cell constant

if $l = 1cm$
$a = 1cm$

then $\boxed{K=G}$ so we can say that conductance of $1cm^3/1ml$ solution is called as specific conductance
or
conductivity

2.C. EQUIVALENT CONDUCTANCE [λ_{eq} OR Λ_{eq}]

Conductance of all the ions produced by **1 gm equivalent** of an electrolyte **in given Volume** of solution is called as equivalent conductance

∵ conductance of 1 cm^3 solution = K

∴ conductance of V cm^3 solution = K × V

$$\boxed{\lambda_{eq} = K \times V}$$

↓ Volm of solution (ml) which contain **1 gm eq** of an electrolyte

Suppose concentration is N eq / lit

∵ Volm of solution containing N gm eq = 1000 ml

∴ Volm of solution Containing 1 gm eq = $\frac{1000}{N}$

$$\lambda_{eq} = K \times V$$

$$\boxed{\lambda_{eq} = K \times \frac{1000}{N}}$$ → Volm of solution (ml) which contain **1 gm eq** of an electrolyte

UNIT [λ_{eq} / Λ_{eq}] = $\Omega^{-1} cm^{-1} \times cm^3 eq^{-1}$

$= \Omega^{-1} cm^2 eq^{-1}$

$= S\ cm^2 eq^{-1}$

2.D- MOLAR CONDUCTANCE [λ_M OR Λ_M OR μ_M]

Conductance of all the ions produced by **1 mol** of an electrolyte in **given volume** of solution is called as MOLAR CONDUCTANCE

$\because$ conductance of $1cm^3$ solution $= K$

$\therefore$ conductance of Vcm^3 solution $= K \times V$

$$\boxed{\lambda_M = K \times V}$$

V → volm of solution (ml) which contain **1 mol** of electrolyte

$\because$ Volume of solution containing M molar $= 1000$ ml

$\therefore$ " " " " 1 molar $= \frac{1000}{M}$ ml

$$\boxed{\lambda_M = K \times V}$$

$$\boxed{\lambda_M = K \times \frac{1000}{M}}$$ → volm of solution (ml) which contain **1 mol** of electrolyte

UNIT [$\lambda_M / \Lambda_M / \mu_M$] $= \Omega^{-1} cm^{-1} \times cm^3 eq^{-1}$

$= \Omega^{-1} cm^2 mol^{-1}$

$= S cm^2 mol^{-1}$

RELATION BETWEEN λ_M AND λ_{eq}

$$\lambda_{eq} = K \times \frac{1000}{N}$$

$$\lambda_{eq} = K \times \frac{1000}{M \times v.f}$$

$$\lambda_{eq} = \frac{\lambda_M}{v.f}$$

$\boxed{\lambda_{eq} \times v.f = \lambda_M}$ Minm. value of v.f 1 तो होगा ही होगा so

$\boxed{\lambda_M \geq \lambda_{eq}}$

conductance and conductivity are additive property जैसे कि अगर कोइ electrolytic solution में several electrolytic ions है तो total conductance of solution willbe

$$C_{TOTAL} = \sum C_{ION} + \sum C_{WATER}$$

FORMULAE

1. $G = \frac{1}{R}$

2. $K = \frac{1}{R} \cdot \frac{l}{a} = G \times G^* = \frac{G}{R}$

3. $\lambda eq = K \times V = K \times \frac{1000}{N}$

4. $\lambda eq = K \times V = K \times \frac{1000}{M}$

5. $\lambda M = \lambda eq \times V \cdot f$

Qsn The resistance of 0.005M K_2SO_4 at 25°C is 200Ω cell constant (K) of conducting cell is 1 cm^{-1} then find equivalent conductance

Ans $\lambda eq = K \times \frac{1000}{N}$

$= K \times \frac{1000}{M \times v.f}$

$= \frac{1}{R} \times G^* \times \frac{1000}{M \times v.f}$

$= \frac{1}{200} \times (1) \times \frac{1000}{0.005 \times 2}$

$= \underline{500\ \Omega^{-1} cm^2 equi^{-1}}$

Qsn Resistance of Decinormal solution present between two platinum electrode that are 1.8 cm apart dipped and 5.4 cm^2 are found to be 50Ω then Calculate λeq of solution

Ans $\lambda eq = K \times \frac{1000}{N}$

$= \frac{1}{R} \times \frac{l}{a} \times \frac{1000}{N}$

$= \frac{1}{50} \times \frac{1.8}{5.4} \times \frac{1000}{1/10}$

$= 66.67\ \Omega^{-1} cm^2 eq^{-1}$

Qsn HOME WORK

0.5 N $Al_2(SO_4)_3$ solution is present between two Platinum electrode that are 2cm x 4cm the resistance of solution is 200Ω. then calculate molar conductance of solution

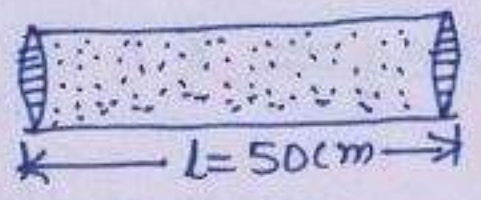
Hints $Al_2SO_4 \longrightarrow 2Al^{+3} + 3SO_4^{-2}$

$+6 \quad -6$

$\lambda M = \frac{K \times 1000 \times v.f}{N}$

Qsn Resistance of a coloumn of 0.5 mole per litre NaOH solution of diametre 1cm and length 50cm is $5.5 \times 10^3 \Omega$. calculate its Resistivity, conductivity and molar conductivity

Ans $R = \rho \cdot \frac{l}{A}$



$$\rho = \frac{RA}{l} = \frac{R(\pi r^2)}{l}$$

$$\rho = \frac{5.5 \times 10^3 \times 3.14 \times (0.5)^2}{50}$$

$$\boxed{\rho = 87.135\ \Omega\text{-cm}}$$

$$k = \frac{1}{\rho} = \frac{1}{87.135\ \Omega\text{-cm}} = 0.01148\ \Omega^{-1}cm^{-1}$$

$$\Lambda_M = k \times \frac{1000}{M} = 0.01148\ \Omega^{-1}cm^{-1} \times \frac{1000}{0.5}$$

OPTION में ये तीनो में से कोई एक रहेगा. जैसे नीचे दिया है।

$$\Lambda_M = 229.6\ \Omega^{-1}(cm^2)\ mol^{-1}$$
$$= 229.6\ \Omega^{-1}(10^{-4}m^2)\ mol^{-1}$$
$$= 229.6 \times 10^{-4}\ \Omega^{-1}m^2\ mol^{-1}$$

(i) $\Lambda_M = 229.6\ \Omega^{-1}(cm^2)\ mol^{-1}$

(ii) $\Lambda_M = 229.6\ \Omega^{-1}(10^{-4}m^2)\ mol^{-1}$

(iii) $\Lambda_M = 229.6 \times 10^{-4}\ \Omega^{-1}m^2\ mol^{-1}$

(iv) All are correct

Ans = (iv)

Qsn Equivalent conductance of 0.1M solution of $MgCl_2$ is 100 mho $cm^2 eq^{-1}$ at 25°c, A cell with electrode that are 1.5 cm is surface area and 0.5 cm apart is filled with 0.1N $MgCl_2$ solution. How much current will flow when the potential difference between electrode is 5V.

Ans $V = IR \qquad I = \frac{V}{R}$ [V तो है But R find करना है]

$$\Lambda eq = K \times \frac{1000}{N}$$

$$100 = \frac{1}{R} \times \frac{l}{a} \times \frac{1000}{N}$$

$$100 = \frac{1}{R} \times \frac{0.5}{1.5} \times \frac{1000}{0.1}$$

$$\boxed{R = \frac{100}{3}} \qquad I = \frac{V}{R} = \frac{5 \times 3}{100} = \frac{15}{100} = \boxed{0.15 \text{ Amp}}$$

Qsn The conductivity of 0.1N NaOH solution is $0.022\ Scm^{-1}$. when equal volume of 0.1N HCl solution is added. The conductivity of resultant solution decreases to $0.0055\ Scm^{-1}$. then calculate equivalent conductivity of Resultant solution

Ans

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

	0.1N	0.1N	
Let vol^m =	1Lit	1Lit	
gm eq =	NXV	NXV	
	= 0.1 gmeq	0.1 gmeq	
After Reaction =	0	0	0.1 gm eq

$$N_{NaCl} = \frac{0.1 \text{ gmeq}}{(1+1) \text{ Lit}}$$

$$= 0.05N$$

$$\Lambda eq = K \times \frac{1000}{N}$$

$$\Lambda eq = 0.0055 \times \frac{1000}{0.050}$$

$$\boxed{\Lambda eq = 110}$$

Qsn HOME WORK When a certain conductance was filled with 0.05 M KCl solution it had resistance of 400 Ω at 25°C when it is filled with $CaCl_2$ solution (11.1 gm in 500 mL solution). it had resistance of 900 Ω if conductivity of 0.05 N KCl solution is 0.00189 S cm^{-1} calculate (Ans = 0.756

(a) Cell constant (Ans = 0.756)

(b) Conductivity of $CaCl_2$ solution (Ans = 8.4×10^{-4} S cm^{-1})

(c) Molar Conductivity of $CaCl_2$ solution (Ans = 4.2 S $cm^2 mol^{-1}$)

(d) equivalent Conductivity of $CaCl_2$ solution (Ans = 2.1 S

FACTOR EFFECTING ELECTROLYTIC CONDUCTANCE

[I] INTERIONIC INTERACTION
[II] SOLVENT OR HYDRATION OF IONS
[III] VISCOCITY
[IV] TEMPERATURE
[V] DILUTION

[I] INTERIONIC INTERACTION

If interionic i is more then mobility of ions will be decreases and thus conductance will decreases

[II] SOLVATION OR HYDRATION

* Solvent molecule का ION से जुड़ना solvation कहलाता है। अब जब ION को solvent पकड़ कर रखेगा तो ION move ही नही कर पाएगा या अगर move करेगा तो slowly move करेगा so conductance decreases.
* जो ion जितना छोटा होता है पानी मे उतना ही ज्यादा मोटा हो जाता है पानी पीकर नही But H_2O को जोड़कर
* SIZE OF ION = $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
* HYDRATED SIZE = $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

[III] VISCOSITY

VISCOSITY बढ़ेगा तो IONS easily move नही कर पाएगा
So VISCOCITY OF MEDIUM INCreases, conductance decreases

[IV] TEMPERATURE

on increasing Temp K.E of IONS INCreases and force of attraction become weak therefore mobility of ions increases. thats why -

Temperature increases (T)↑ = conductance (G)↑

[iv] DILUTION

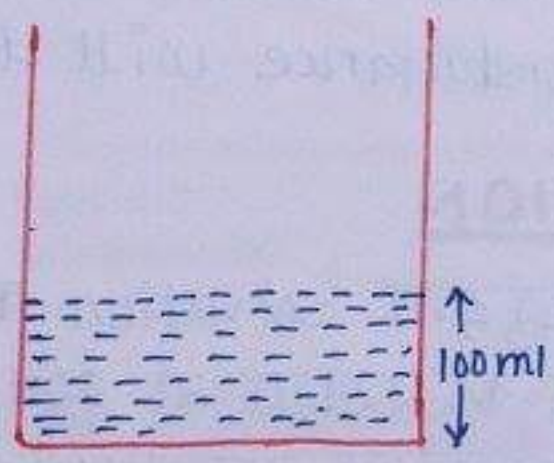
अब तक जो तीन effect था Interionic interaction, solvation and temperature के तीनो common concept था. But effect of dilution on conductance is so important concept.

DILUTION INCREASES = CONDUCTANCE INCREASES (G)

DILUTION INCREASES = CONDUCTIVITY DECRESES (K)

BEFORE DILUTION

1 mol present in 100 ml



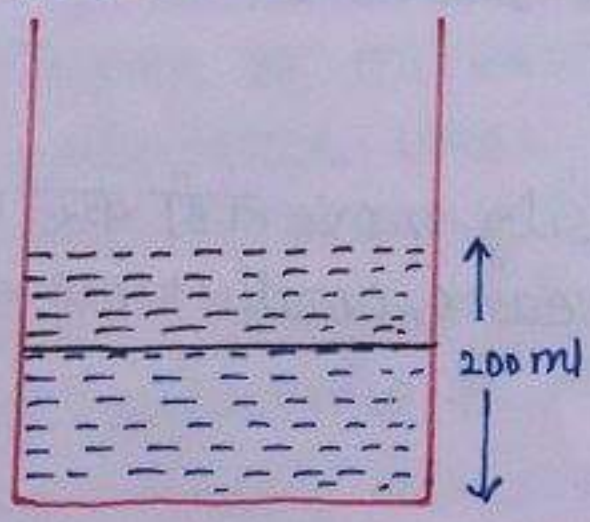
Let

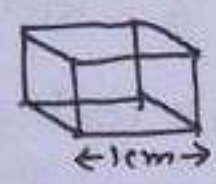
100 ml soln have = 50,00000 ions = $G_1 = \wedge M_1$

1 ml soln have = 50,00000 ions = K_1 [K Always 1ml के लिए होता है]

AFTER DILUTION

1 mol in 200 ml



1 ml = 

$G_2 > G_1$

$K_1 > K_2$

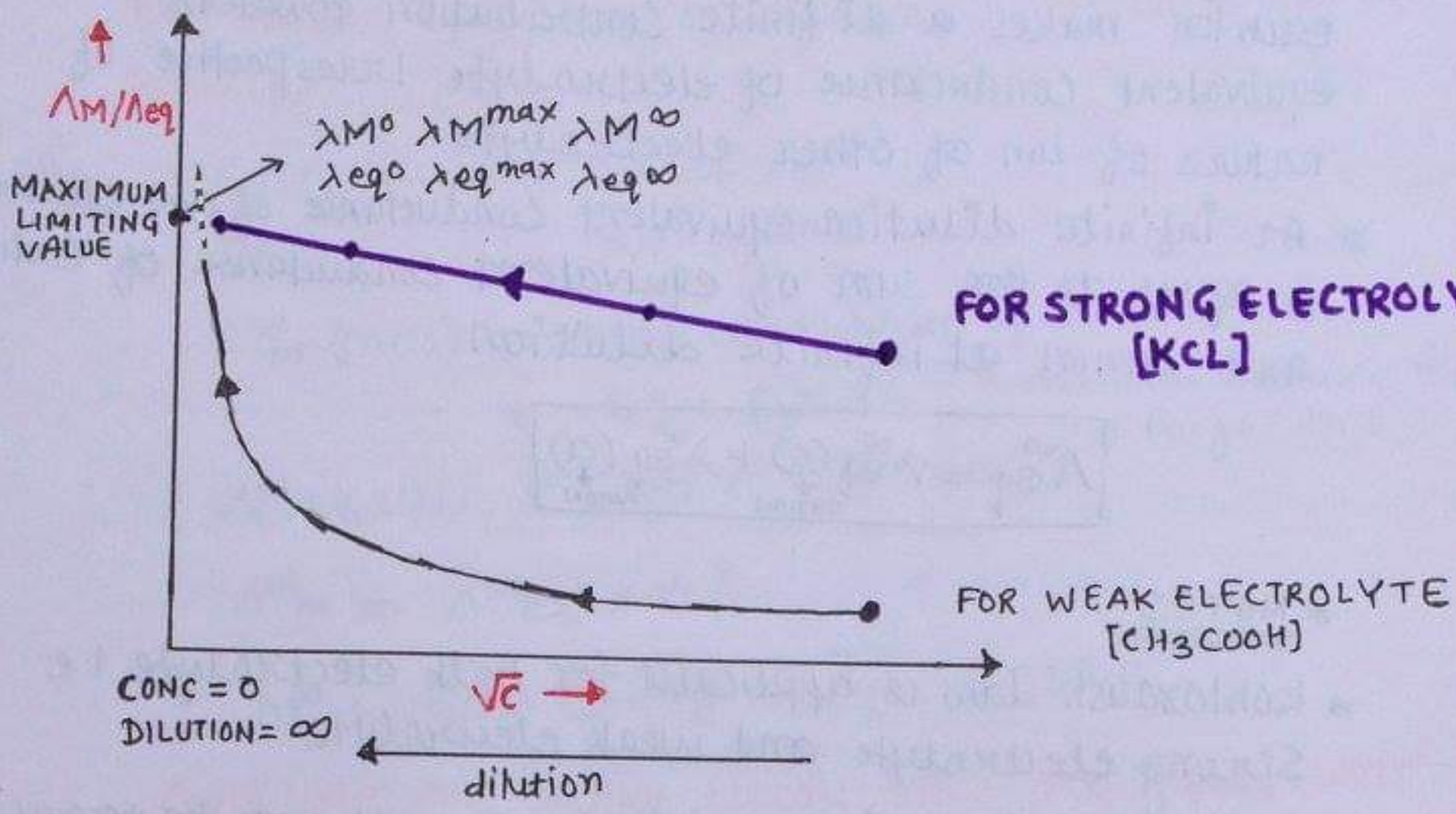
200 ml soln contains 60,00000 ions = $G_2 = \wedge M_2$

1 ml soln contains 30.00000 ions = K_2

* ON DILUTION ⇒ G ↑ BUT K ↓

* ON DILUTION ⇒ $\wedge eq$ ↑ AND $\wedge M$ ↑ [दोनो INCREASES]

DEBYE-HUCKEL-ONSAGAR GRAPH



FOR STRONG ELECTROLYTE

$$\Lambda m = \Lambda M^{\infty} - b\sqrt{c}$$

OR

$$\Lambda eq = \Lambda eq^{\infty} - b\sqrt{c}$$

→ DEBY-HUCKLE-ONSAGAR EQUATION

→ b = DHO constant

→ b depends upon - Nature of electrolyte
- Nature of solvent
- Temperature

OBSERVATION OF DHO-GRAPH

* Dilution को increase करने पर strong electrolyte की molar Conductance weak electrolyte की molar conductance के जितना increase नही होता.
BUT strong electrolyte की Conductance तो weak electrolyte की Conductance से Always greater ही होगा.
* on increase dilution conduction of strong electrolyte तो gradually increase होती है। But conduction of weak electrolyte rapidly increase करती है।
* अगर Graph को extraplot किया जाए तो strong electrolyte की λM obtain किया जा सकता है। But weak electrolyte की maximum λM obtain नही किया जा सकता है, क्यो कि weak electrolyte की α is always less than 1

KOHLRAUSCH'S LAW

* At infinite dilution when dissociation is completed each ion makes a difinite contribution towards equivalent conductance of electrolyte irrespective of nature of ion of other electrolyte
* At infinite dilution equivalent conductance of an electrolyte is equal to the sum of equivalent conductance of cation and anion at infinite dilution

$$\boxed{\Lambda^{\infty}_{eq} = \lambda^{\infty}_{eq}(c)_{\text{CATION}} + \lambda^{\infty}_{eq}(a)_{\text{ANION}}}$$

* NOTES

⇒ kohlaraush law is applicable for both electrolyte i.e strong electrolyte and weak electrolyte

⇒ kohlraush has given original statement was in terms of equivalent conductance but according to I.U.P.A.C convention now kohlraush law is expressed interms of molar conductance

$$\boxed{\Lambda^{\infty}_{M} = m\,\lambda^{\infty}_{M}(c) + n\,\lambda^{\infty}_{M}(a)}$$

m = NO of moles of cation in 1 mol of electrolyte
n = NO of moles of anion in 1 mol of electrolyte

eg:- $K_2SO_4 \longrightarrow 2K^{+} + SO_4^{-2}$

1 mol — 2 mol — 1 mol
m = 2 — n = 1
1 gm eq — 1 gm eq + 1 gm eq

eg:- $Al_2(SO_4)_3 \longrightarrow 2Al^{+} + 3SO_4^{2-}$

1 mol — 2 mol — 3 mol
m = 2 — n = 3
1 gm eq — 1 gm eq + 1 gm eq

Qsn ionic conductance at infinite dilution of Ca^{2+} and Cl^- are 119 and 76 $cm^2 mol^{-1}$ respectively. Calculate the molar equivalent conductance of $CaCl_2$ at infinite dilution

Ans

$$CaCl_2 \longrightarrow Ca^{+2} + 2Cl^-$$
1mol 1mol 1mol

$\Lambda^{\infty}_M (CaCl_2)$ = molar conductance of Cation + Anion

$= 119 + (2) \times 76$ → maximum mistake we forget this

$\Lambda^{\infty}_M (CaCl_2) = 271\ \Omega^{-1} cm^2 mol^{-1}$

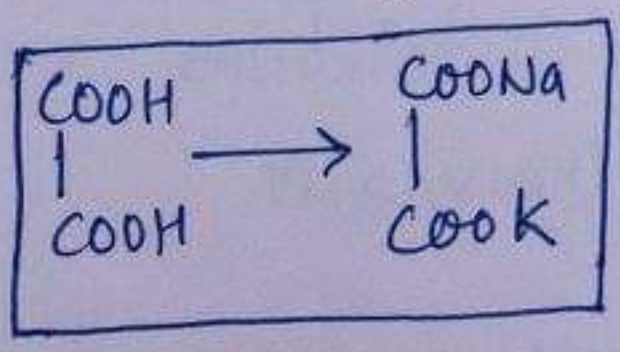
$\Lambda^{\infty}_M = \Lambda^{\infty}_{eq} \times V.f$

$\Lambda^{\infty}_{eq} (CaCl_2) = \frac{\Lambda^{\infty}_M}{V.f} = \frac{271}{2} = 135.5$

Qsn Calculate the molar conductance and equivalent conductance at infinite dilution of the salt KOOC.COONa. Given the ionic conductance is at infinite dilution O^{2-} (oxalate ion), K^+ and Na^+ are 148.2, 50.1 and 73.5 $\Omega^{-1} cm^2 mol^{-1}$

Ans KOOC.COONa ⟶ $C_2O_4^{2-} + Na^+ K^+$

[Sodium potassium oxalate] 1mol 1mol 1mol



$\Lambda^{\infty}_M = \lambda^{\infty}_M (OX^{-2}) + \lambda^{\infty}_M Na^+ + \lambda^{\infty}_M K^+$

$\Lambda^{\infty}_M = 148.2 + 50.1 + 73.5$

$\Lambda^{\infty}_M = 271.8\ \Omega^{-1} cm^2 mol^{-1}$

$\Lambda^{\infty}_{eq} = \frac{\Lambda^{\infty}_M}{V.f} = \frac{271.8}{2} = 135.9$

APPLICATION OF KOHLRAUSH LAW

[I] To Determine degree of ionisation (α)
[II] To determine dissociation constant [K]
(III) To determine Λ_M^∞ OR λ_{eq}^∞ for weak electrolyte
[iv] To determine solubility of sparingly soluble salt

[I] TO DETERMINE DEGREE OF IONISATION

$$\alpha = \frac{\text{Dissociated moles}}{\text{Total moles}} = \frac{\text{ions obtained by dissociation of electrolyte}}{\text{Total ions present in electrolyte}}$$

At constan "concentration c"

$$\alpha = \frac{\Lambda_{eq}^c}{\Lambda_{eq}^\infty} \text{ or } \frac{\Lambda_M^c}{\Lambda_M^\infty}$$

[II] TO DETERMINE DISSOCIATION CONSTANT [K]

eg- for CH_3COOH

$$Ka = \frac{c\alpha^2}{(1-\alpha)} \quad \text{if } \alpha << 1 \text{ then } Ka = c\alpha^2$$

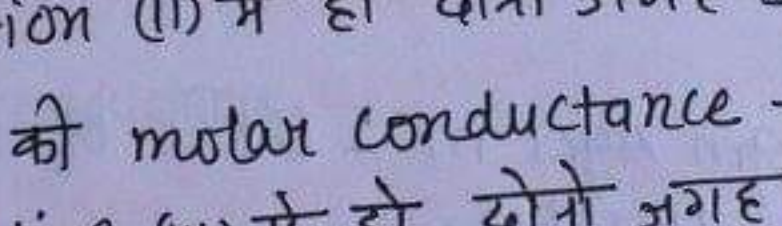
$$[H^+] = c\alpha$$

$$i = 1 - \alpha + n\alpha$$

[III] TO DETERMINE Λ_M^∞ OR Λ_{eq}^∞ FOR A WEAK ELECTROLYTE

QSN: $\Lambda^\infty_M (CH_3COOH) = ?$

* यहाँ CH_3COOH एक weak electrolyte है। और graph से weak electrolyte का $\Lambda^\infty M$ find नहीं कर सकते हैं। Graph से तो सिर्फ strong electrolyte का ही $\Lambda^\infty M$ find कर सकते



* But we can use Kohlraush law because Kohlraush law is applicable for both strong and weak electrolyte

$$\Lambda_M^\infty (CH_3COOH) = \lambda^\infty_M (CH_3COO^-) + \lambda_M^\infty (H^+) \quad ----(i)$$

$$\Lambda_M^\infty (CH_3COONa) = \lambda^\infty_M (CH_3COO^-) + \lambda^\infty_M (Na^+) \quad ----(ii)$$

$$\Lambda^\infty_M (HCl) = \lambda^\infty_M (H^+) + \lambda^\infty_M (Cl^-) \quad ----(iii)$$

$$\Lambda^\infty_M (NaCl) = \lambda^\infty_M (Na^+) + \lambda^\infty_M (Cl^-) \quad ----(iv)$$

* $\lambda^\infty_M (CH_3COO^-)$ की Molar conductance चाहे equation (i) में हो या equation (ii) में हो दोनों जगह same ही होगी

* $\lambda^\infty_M (H^+)$ की molar conductance चाहे equation (i) में हो या equation (iii) में हो दोनों जगह same ही होगी

* equation (i) = equation (ii) + equation (iii) − equation (iv)

$$\Lambda^\infty_M (CH_3COOH) = \Lambda^\infty_M (CH_3COO^-) + \lambda^\infty_M (H^+) - \Lambda^\infty_M (NaCl)$$

$$\Lambda^\infty_M (CH_3COOH) = \Lambda^\infty_M (\underline{CH_3COO}Na + \underline{H}Cl + NaCl)$$

[IV] TO DETERMINE SOLUBILITY OF SPARINGLY SOLUBLE SALT

Eg- AlCl, $BaSO_4$, $PbSO_4$

Solubility

- Concentration of saturated solution at constant temperature called solubility
- The maximum amount of a substance which can be dissolve in 100 gm of solvent to prepare its saturated solution

FUNDA:

- saturated solution of sparingly soluble salt can be considered as infinitely diluted solution because these salt dissolve very less therefore

saturated solution = infinitely diluted solution

$$\Lambda_M = \Lambda^{\infty}_M = K \times \frac{1000}{M}$$

By Kohlraush law

यहाँ पर M=S

*if we add 500 gm AgCl in 1 litre of H_2O then only upto 5g AgCl is dissolve.
*495 gm AgCl remain undissolve. so actual solubility = 5g/L

Qsn At a certain temperature the saturated solution of AgCl has conductivity (K) 1.8×10^{-6} mho cm^{-1}. The ionic conductance is of Ag^+ and Cl^- at infinite dilution are 54.5 and 65.5 mho $cm^2 mol^{-1}$. Find the solubility of AgCl at this temperature

Ans $\Lambda^{\infty}_M AgCl = 54.5 + 65.5$

$= 120$

$$\Lambda^{\infty}_M = 1.8\times10^{-6}\times\frac{1000}{S}$$

$$120 = 1.8\times10^{-6}\times\frac{1000}{S}$$

$$S = \frac{1.8\times10^{-6}\times1000}{120}$$

$S = 1.5\times10^{-5}$ mol/litre

$$Ksp = S^2 = 2.25\times10^{-10}$$

Qsn The molar conductanie for acetic acid at 0.2M Molar conductance at infinite dilute are 4.8 and 400 $\Omega^{-1} cm^2 mol^{-1}$ respectively. Calculate degree of dissociation and dissociation constant of CH_3COOH.

Ans $C = 0.2\,M\ CH_3COOH$

$\Lambda^{c}_M = 4.8$

$\Lambda^{\infty}_M = 400\ \Omega^{-1} cm^2 mol^{-1}$

$$\alpha = \frac{\Lambda^{c}_M}{\Lambda^{\infty}_M} = \frac{4.8}{400}\times100 = 1.2\% = 1.2\times10^{-2}$$

$Ka = c\alpha^2$

$= 0.2\,(1.2\times10^{-2})^2$

$\simeq 2.88\times10^{-5}$

Qsn The molar conductance of an infinitely diluted solution of Ammonium Chloride (NH_4Cl) is $150\,\Omega^{-1}cm^2mol^{-1}$ and ionic conductance is ($OH^- = 198$), ($Cl^- = 76\,\Omega^{-1}cm^2mol^{-1}$) What will be the molar conductance of the solution of NH_4OH (Ammonium Hydroxide) at infinite dilution. If the conductivity of 0.01M solution of NH_4OH is $9.52 \times 10^{-5}\,\Omega^{-1}cm^2$ what will be its degree of dissociation

Ans $NH_4Cl = 150\ Ohm^{-1}\,cm^2\,mol^{-1}$

$OH^- = 198$

$Cl^- = 76\,\Omega^{-1}cm^2\,mol^{-1}$

$\Lambda^\infty_M(NH_4OH) = \Lambda^\infty_M(NH_4Cl + OH^- - Cl^-)$

$\Lambda^\infty M(NH_4OH) = \Lambda^\infty_M(150 + 198 - 76)$

$\Lambda^\infty_M(NH_4OH) = 272\,\Omega^{-1}cm^2mol^{-1}$

$C = 0.01M$

$K = 9.52 \times 10^{-5}\,\Omega^{-1}cm^{-1}$

$\Lambda^c_M = \dfrac{K \times 1000}{M}$

$\Lambda^cM = \dfrac{9.52 \times 10^{-5} \times 1000}{0.01} = 9.52$

$\alpha = \dfrac{\Lambda^cM}{\Lambda^\infty_M} = \dfrac{9.52}{272} \times 100$

$\alpha = 3.5\%$

Qsn Conductivity of saturated solution of AgCl is $2.2\times10^{-6}\,\Omega^{-1}cm^{-1}$ and conductivity of water is $0.8\times10^{-6}\,\Omega^{-1}cm^{-1}$. Molar conductance at infinite dilution of $AgNO_3$, HCl and HNO_3 are 135, 426 and 421 $\Omega^{-1}cm^2mol^{-1}$ respectiv Calculate solubility product of AgCl

Ans $AgNO_3 = 135$

$HCl = 426$

$HNO_3 = 421$

$$\Lambda_M^\infty(AgCl) = \Lambda_M^\infty(AgNO_3 + HCl - HNO_3)$$

$$= \Lambda_M^\infty(135 + 426 - 421)$$

$$\Lambda_M^\infty(AgCl) = 140\,\Omega^{-1}cm^2mol^{-1}$$

$K_{AgCl\ solution} = 2.2\times10^{-6}\,\Omega^{-1}cm^{-1}$

$K_{H_2O\ solution} = 0.8\times10^{-6}\,\Omega^{-1}cm^{-1}$

It should neglect

$K_{solution\ of\ electrolyte} = K_{electrolyte} + K_{H_2O}$

$= K_{solute\ electrolyte} + K_{H_2O}$

$= 2.2\times10^{-6} - 0.8\times10^{-6}$

$K_{AgCl} = 1.4\times10^{-6}$

Qsn specific conductance of 0.03M aqua solution of weak base BOH is $1.5\times10^{-4}\,Scm^{-1}$. also the molar conductance at infinite dilution is $250\,Scm^2mol^{-1}$ find ionisation constant of weak base

Ans $C = 0.03M$

$K = 1.5\times10^{-4}\,Scm^{-1}$

$\Lambda_M^\infty = 250\,Scm^2mol^{-1}$

$$\Lambda_M^C = K\times\frac{1000}{M}$$

$$\Lambda_M^C = 1.5\times10^{-4}\times\frac{1000}{0.03}$$

$$\Lambda_M^C = 5$$

$$\alpha = \frac{\Lambda_M^C}{\Lambda_M^\infty} = \frac{5}{250} = 0.02$$

$$K_b = \frac{C\alpha^2}{1-\alpha} = C\alpha^2$$

$$K_b = 0.03\,(0.02)^2$$

$$K_b = 1.2\times10^{-5}$$

AIIMS

ELECTROLYTE	KCl	KNO_3	HCl	NaOAc	NaCl
Λ^∞ (Scm^2mol^{-1})	149.9	145.0	426.2	91.0	126.5

Calculate Λ^∞_{HOAc} using appropriate molar conductance of electrolyte listed above at infinite dilution in H_2O at 25°C

Ans $$\Lambda^\infty_{HOAc} = \Lambda^\infty_{NaOAc} + \Lambda^\infty_{HCl} - \Lambda^\infty_{NaCl}$$

$$= 91.0 + 426.2 - 126.5 \; Scm^2mol^{-1}$$

$$= 390.7 \; Scm^2mol^{-1}$$

Qsn AIIMS Hydrofluoride acid is weak acid. At 25°C the molar conductivity of 0.002 M HF is $176.2 \Omega^{-1}cm^2mol^{-1}$ if its $\Lambda^\circ_m = 405.2 \Omega^{-1} cm^2 mol^{-1}$. Equilibrium constant at the given concentration is

Ans $$\alpha = \frac{\Lambda_c}{\Lambda^\infty_m} = \frac{176.2 \Omega^{-1} cm^2 mol^{-1}}{405.2 \Omega^{-1} cm^2 mol^{-1}} = 0.435$$

$$K = \frac{[H^+][F^-]}{[HF]} = \frac{c\alpha^2}{1-\alpha} = \frac{(0.002M)(0.435)^2}{1-0.435}$$

$$K = 6.70 \times 10^{-4} M$$